

Department of Computer Science and Engineering

PES UNIVERSITY

UE19CS251: Design and Analysis of Algorithm (4-0-0-4-4)

Q.1 what does dynamic programming have in common with divide-and conquer?

Solution:

- a. Both techniques are based on dividing a problem's instance into smaller instances of the same problem.
- b. Typically, divide-and-conquer divides an instance into smaller instances with no intersection whereas dynamic programming deals with problems in which smaller instances overlap. Consequently, divide-and-conquer algorithms do not explicitly store solutions to smaller instances and dynamic programming algorithms do.

Q.2 Write pseudocode of the bottom-up dynamic programming algorithm for the knapsack problem

Solution:

Bottom up Dynamic knapsack 0/1 ($v[1..n]$, $w[1..n]$, n , W)

for $j = 0$ to W do

$\text{cost}[0, w] = 0$

for $i = 1$ to n do

$\text{cost}[i, 0] = 0$

 for $j = 1$ to W do

 if $w_i \leq j$ then

 if $v_i + \text{cost}[i-1, w-w_i]$ then

$\text{cost}[i, w] = v_i + \text{cost}[i-1, w-w_i]$

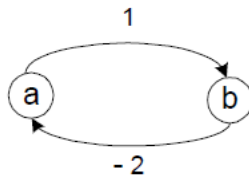
 else $\text{cost}[i, w] = \text{cost}[i-1, w]$

 else

$\text{cost}[i, w] = \text{cost}[i-1, w]$

Q.3 Give an example of a graph with negative weights for which Floyd's algorithm does not yield the correct result

Solution:



Floyd's algorithm will yield:

$$D^{(0)} = \begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix} \quad D^{(1)} = \begin{bmatrix} 0 & 1 \\ -2 & -1 \end{bmatrix} \quad D^{(2)} = \begin{bmatrix} -1 & 0 \\ -3 & -2 \end{bmatrix}$$

None of the four elements of the last matrix gives the correct value of the shortest path, which is, in fact, $-\infty$ because repeating the cycle enough times makes the length of a path arbitrarily small.

Q.4 Enhance Floyd's algorithm so that shortest paths themselves, not just their lengths, can be found

Solution:

Algorithm *FloydEnhanced*($W[1..n, 1..n]$)

//Input: The weight matrix W of a graph or a digraph

//Output: The distance matrix $D[1..n, 1..n]$ and

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//          the matrix of intermediate updates  $P[1..n, 1..n]$ 
 $D \leftarrow W$ 
for  $i \leftarrow 1$  to  $n$  do
  for  $j \leftarrow 1$  to  $n$  do
     $P[i, j] \leftarrow 0$  //initial mark
for  $k \leftarrow 1$  to  $n$  do
  for  $i \leftarrow 1$  to  $n$  do
    for  $j \leftarrow 1$  to  $n$  do
      if  $D[i, k] + D[k, j] < D[i, j]$ 
         $D[i, j] \leftarrow D[i, k] + D[k, j]$ 
         $P[i, j] \leftarrow k$ 
  
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